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Achievable Error Exponents for the Private Fingerprinting Game

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Abstract

Fingerprinting systems in the presence of *collusive* attacks are analyzed as a game between a fingerprinter and a decoder, on the one hand, and a coalition of two or more attackers, on the other hand. The fingerprinter distributes, to different users, different fingerprinted copies of a host data (*covertext*), drawn from a memoryless stationary source, embedded with different fingerprints. The coalition members create a forgery of the data while aiming at erasing the fingerprints in order not to be detected. Their action is modelled by a multiple access channel (MAC). We analyze the performance of two classes of decoders, associated with different kinds of error events. The decoder of the first class aims at detecting the *entire* coalition, whereas the second is satisfied with the detection of *at least one* member of the coalition. Both decoders have access to the original covertext data and observe the forgery in order to identify member/s of the coalition. Motivated by a worst-case approach, we assume that the coalition of attackers is informed of the hiding strategy taken by the fingerprinter and the decoder, while they are uninformed of the attacking scheme. Single letter expressions for the error exponents of the two kinds are obtained, a decoder that is optimal with respect to the two kinds of errors is introduced, and the worst-case attack channel is characterized.

1 Introduction

In fingerprinting systems several copies of the same host data are embedded with different fingerprints (that designate, e.g., the different digital signatures or serial numbers of the copies they are provided with) and distributed to different users . The fingerprints identify one of many users in order to enable copyright protection. In this situation, two or more users can form a coalition, and collusive attacks on the fingerprinting system are possible and have to be taken into account in the code design. Each of the coalition members contributes his distinct fingerprinted copy in order to create a better forgery. Hence, the fingerprinting problem can be thought of as a game between the fingerprinter and the coalition of attackers.